

Assignment 2

Assignment Topics:

Local Searches, Adversarial Searches, Constraint specification problem and Propositional logics

Guidelines:

1. **Deadlines should be kept in mind. No extension in assignment dates would be given. No late submission will be accepted.**
2. **This is an individual assignment. PLAGIARISM IS NOT ACCEPTABLE AT ALL! Zero marks will be given in case of plagiarism.**
3. **Deadline = 18-07-2025, Deadline is hard and firm.**

Question No 01:

Which of the following are correct? Validate your answers using truth-table.

- a) $\text{False} \models \text{True}$.
- b) $\text{True} \models \text{False}$.
- c) $(A \wedge B) \models (A \Leftrightarrow B)$.
- d) $A \Leftrightarrow B \models A \vee B$.
- e) $A \Leftrightarrow B \models \neg A \vee B$.
- f) $(A \wedge B) \Rightarrow C \models (A \Rightarrow C) \vee (B \Rightarrow C)$.
- g) $(C \vee (\neg A \wedge \neg B)) \equiv ((A \Rightarrow C) \wedge (B \Rightarrow C))$.
- h) $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B)$.
- i) $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B) \wedge (\neg D \vee E)$.
- j) $(A \vee B) \wedge \neg(A \Rightarrow B)$ is satisfiable.
- k) $(A \Leftrightarrow B) \wedge (\neg A \vee B)$ is satisfiable.
- l) $(A \Leftrightarrow B) \Leftrightarrow C$ has the same number of models as $(A \Leftrightarrow B)$ for any fixed set of proposition symbols that includes A, B, C.

Question No 02:

Consider the problem of placing k queen on an $n \times n$ chessboard such that no two queens are attacking each other, where k is given and $k \leq n^2$

1. Choose a CSP formulation. In your formulation, what are the variables?
2. What are the possible values of each variable?
3. What sets of variables are constrained, and how?

Question no 03:

A turn-based strategy game between two players uses a minimax strategy with alpha-beta pruning. The current game tree has:

- Branching factor = 4
- Depth = 6

What is the maximum number of nodes that would be evaluated without pruning?

What is the minimum number of nodes that could be evaluated with perfect pruning?

Explain the conditions necessary to achieve this optimal pruning.

Question No 04:

Design your own original minimax game tree with:

- At least 3 levels (Max-Min-Max)
 - At least 12 leaf nodes
 - Include at least one pruning opportunity
1. Draw the tree.
 2. Assign utility values to leaf nodes.
 3. Perform alpha-beta pruning with clear indication of:
 - α and β updates
 - Pruned branches
 4. State the final decision at the root.

Question No 05:

Consider a multi-peaked landscape (fitness landscape) with several local maxima. Explain how simple hill climbing can get stuck in a local maximum. Then, provide a scenario where:

1. Greedy hill climbing fails to reach the global maximum.
2. Random-restart hill climbing succeeds where greedy fails.
3. Illustrate with a diagram and label the local and global maxima.

Question No 06:

The Chrono-Science Enthusiasts Society (CSES) at a prestigious university is hosting a symposium featuring seven distinguished scientists from history, each of whom has traveled through time to deliver a talk. The organizers have only four one-hour time blocks available (from 1pm to 4pm), and you've been assigned the task of organizing the lecture schedule.

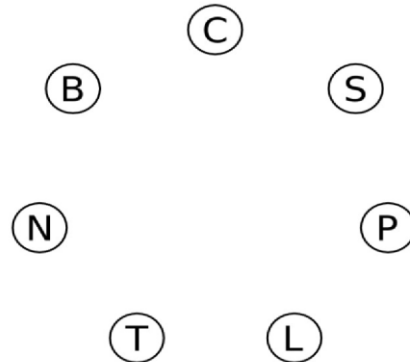
However, based on feedback from attendees, certain groups of students wish to attend multiple specific talks and would be disappointed if those lectures overlap. For example, the physics interest group insists on attending both Niels Bohr and Isaac Newton's sessions, so their talks must be in different time slots.

After conducting a survey, you've categorized student preferences into several groups. Each group lists the talks they wish to attend, and each student belongs to at most one group.

The list of guest lecturers consists of Alan **T**uring, Ada **L**ovelace, Niels **B**ohr, Marie **C**urie, **S**ocrates, **P**ythagoas, and Isaac **N**ewton.

1. Turing has to get home early to help win World War II, so he can only be assigned to the 1pm slot.
2. The Course VIII students want to see the physicists: Bohr, Curie, and Newton.

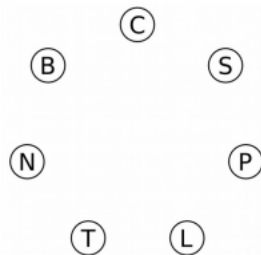
3. The Course XVIII students want to see the mathematicians: Lovelace, Pythagoras, and Newton.
4. The members of the Ancient Greece Club wants to see the ancient Greeks: Socrates and Pythagoras.
5. The visiting Wellesley students want to see the female speakers: Lovelace and Curie.
6. The CME students want to see the British speakers: Turing, Lovelace, and Newton.
7. Finally, you decide that you will be happy if and only if you get to see both Curie and Pythagoras. (Yes, even if you belong to one or more of the groups above.)



Part A: Diagram these constraints by drawing a line between the initials of each pair of guests who cannot share a time slot.

Part B: Search for a solution using depth-first search only—without any forward checking or propagation. The only check is to make sure that each new assignment violates no constraint with any previous assignment. As a tiebreaker, assign a lecturer to the earliest available timeslot. Continue up to the first time you try and fail to assign any time to Newton and must backtrack, at which point you give up and move on to Part C to try a more sophisticated approach. Show your answers on the next two pages

Constraint graph for this problem



Domains for this problem

T	1
L	1 2 3 4
B	1 2 3 4
C	1 2 3 4
S	1 2 3 4
P	1 2 3 4
N	1 2 3 4

Fill out this worksheet as you draw your search tree. There may be more rows than you need. 1. Every time you assign a variable or remove a variable from the propagation queue, fill out a new row in the table. (The same variable might appear in more than one row, especially if you have to backtrack.) 2. In that row, indicate which variable you assigned or de-queued; write its assigned value if it has one (e.g. $X=x$), otherwise just write its name (X). In the second column, list the values that were just eliminated from neighboring variables as a result. If no values were just eliminated, write NONE instead. 3. If your search has to backtrack after assigning or de-queuing a variable: first, finish listing all values eliminated from neighboring variables in the current row. Next, check the backtrack box in that row. Then, continue with the next assignment in the following row as usual. 4. At some point, you might add several variables to your propagation queue at once. Break ties by adding variables to your propagation queue in alphabetical order.

T
L
B
C
S
P
N

	Var assigned or de- queued	List all values eliminated from neighboring variables	Back track ?		Var assigned or de-queued	List all values eliminated from neighboring variables	Back track ?
ex	X	Y ≠ 3, 4 Z ≠ 3 (example)	☒	10			☐
1			☐	11			☐
2			☐	12			☐
3			☐	13			☐
4			☐	14			☐
5			☐	15			☐
6			☐	16			☐
7			☐	17			☐
8			☐	18			☐
9			☐	19			☐

Question No 08:

(a) Decide whether each of the following sentences is valid, unsatisfiable, or neither with some proof:

- $(\text{Smoke} \Rightarrow \text{Fire}) \Rightarrow (\neg \text{Smoke} \Rightarrow \neg \text{Fire})$
- $\text{Smoke} \vee \text{Fire} \vee \neg \text{Fire}$
- $((\text{Smoke} \wedge \text{Heat}) \Rightarrow \text{Fire}) \Leftrightarrow ((\text{Smoke} \Rightarrow \text{Fire}) \vee (\text{Heat} \Rightarrow \text{Fire}))$
- $(\text{Smoke} \Rightarrow \text{Fire}) \Rightarrow ((\text{Smoke} \wedge \text{Heat}) \Rightarrow \text{Fire})$
- $\text{Big} \vee \text{Dumb} \vee (\text{Big} \Rightarrow \text{Dumb})$

(b) Consider the following sentence:

$$[(\text{Food} \Rightarrow \text{Party}) \vee (\text{Drinks} \Rightarrow \text{Party})] \Rightarrow [(\text{Food} \wedge \text{Drinks}) \Rightarrow \text{Party}]$$

- Determine, using enumeration, whether this sentence is valid, satisfiable (but not valid), or unsatisfiable.
- Convert the left-hand and right-hand sides of the main implication into CNF, showing each step, and explain how the results confirm your answer to (a).
- Prove your answer to (a) using resolution.

Question no 09:

- Explain the concept of Simulated Annealing in optimization algorithms. How does it emulate the annealing process in metallurgy, and how is temperature (T) used in the algorithm to control the probability of accepting bad moves? Provide a brief explanation with an example.
- In Simulated Annealing, the probability of accepting a move from a state with energy E1 to a state with energy E2. where e denotes the exponential function, k is Boltzmann's constant, and T is the temperature parameter. If E1 = 10, E2 = 5, and T = 2, calculate the probability of accepting this move. Show your calculations.